Mahdi Kallel

PhD candidate in reinforcement learning

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Profile

I am a final-year PhD candidate at the University of Würzburg, advised by Prof. Carlo D'Eramo, and expect to graduate in April 2027. I began my PhD developing policy search and policy optimization methods, and this work led me to ask how the networks inside these algorithms actually learn. My research now focuses on the learning dynamics of neural networks, in particular memory and plasticity in continual learning and how these dynamics differ between reinforcement and supervised learning. Ultimately, I want to turn this understanding into better training methods.

Publications

2026 **Mahdi Kallel**, Nour Jamoussi, Johannes Tölle, Marios Kountouris, Carlo D'Eramo. **The Spectral Anatomy of Plasticity Loss**. *AXIOM Workshop at NeurIPS 2026*.
Uses random matrix theory to show that learned structure persists as plasticity declines, while later parameter updates increasingly concentrate in existing spectral directions.

2026 **Mahdi Kallel**, Johannes Tölle, Ahmed Hendawy, Carlo D'Eramo. **Do Not Imitate, Reinforce: Iterative Classification via Belief Refinement**. *Transactions on Machine Learning Research (TMLR), 2026*.
Trains classifiers to refine their predictions through reinforcement learning, improving the balance between accuracy and calibration compared with cross-entropy. The critic estimates remaining improvement and decides when to stop.

2026 **Mahdi Kallel**, Jose-Luis Holgado-Alvarez, Samuele Tosatto, Carlo D'Eramo. **PPO+: Enhancing Proximal Policy Optimization**. *Reinforcement Learning Journal (RLC 2026)*.
Shows that the critic is a bottleneck in PPO, and that adding off-policy data for critic training improves on-policy gradient estimates and performance.

2026 Ahmed Hendawy, Henrik Metternich, Théo Vincent, **Mahdi Kallel**, Jan Peters, Carlo D'Eramo. **Use the Online Network If You Can: Towards Fast and Stable Reinforcement Learning**. *ICLR 2026*.
Reduces the learning delay of target networks by taking the minimum of online and target value estimates, enabling faster learning while controlling overestimation.

2024 **Mahdi Kallel**, Debabrota Basu, Riad Akrour, Carlo D'Eramo. **Augmented Bayesian Policy Search**. *ICLR 2024*.
Introduces a Gaussian process prior that guides policy search with the policy gradient, enabling Bayesian optimization to scale to high-dimensional control problems.

2024 Oliver Järnefelt, **Mahdi Kallel**, Carlo D'Eramo. **Cyclicity-Regularized Coordination Graphs**. *RLC 2024*.
Shows that accounting for cycles when sparsifying coordination graphs improves joint action selection and performance in cooperative multi-agent reinforcement learning.

Research Experience

2022 **Research Intern**, *INRIA Lille, SCOOL Research Group*, France
- Started research on Bayesian policy search with deterministic policies, leading to Augmented Bayesian Policy Search (ICLR 2024).

2020 **Research Intern**, *EURECOM*, Biot, France
- Compared centralized and decentralized approaches to mobile traffic prediction and load balancing.

Industry Experience

2021 **Machine Learning Engineer**, *Mcovision (independent)*, Paris, France
- Built a classifier of echocardiography videos for cardiac amyloidosis and other rare heart diseases with Hôpital Cochin, from data cleaning to training.

2021 **Research Intern**, *SustainEcho*, Paris, France
- Explored few-shot and hierarchical learning methods for classification with little data.

Education

2023–2027 **PhD, Reinforcement Learning**, *University of Würzburg*, Germany
Expected April 2027. Supervisor: Prof. Carlo D'Eramo.
Teaching: Introduction to Machine Learning; Introduction to Reinforcement Learning.

2021–2022 **MSc, Optimization**, *Université Paris-Saclay*, France
Coursework: Convex analysis, dynamic programming, stochastic optimization and approximation.

2020–2021 **MSc, Mathematics, Vision, Learning (MVA)**, *ENS Paris-Saclay*, France
Coursework: Reinforcement learning, convex optimization, kernel methods, advanced deep learning.

2018–2020 **BSc, Data Science**, *Télécom Paris / EURECOM*, France

Supervision, Talks and Service

Supervision Johannes Tölle, MSc thesis, University of Würzburg (2026). The thesis received the VDE Bayern Award 2026, and Johannes co-authored two of the papers above.

Talks *On Better Priors for Policy Search*. Invited talk and lab visit, Improbable AI Lab (Pulkit Agrawal), MIT, August 2024.

Reviewer NeurIPS (top reviewer, 2026), ICLR, ICML, RLC

Technical Skills

Programming Python, JAX, PyTorch, NumPy, Julia, C

Languages Arabic (native), English (fluent), French (fluent), German (basic)